

6. The age of a person (in years) in 2025 is a perfect square. His age (in years) was also a perfect square in 2012. His age (in years) will be a perfect cube m years after 2025. Determine the smallest value of m .
7. The sum of two real numbers is a positive integer n and the sum of their squares is $n + 1012$. Find the maximum possible value of n .
8. A quadrilateral has four vertices A, B, C, D . We want to colour each vertex in one of the four colours red, blue, green or yellow, so that every side of the quadrilateral and the diagonal AC have end points of different colours. In how many ways can we do this?
9. Four sides and a diagonal of a quadrilateral are of lengths 10, 20, 28, 50, 75, not necessarily in that order. Which amongst them is the only possible length of the diagonal?

Space for rough work